

SIMATS ENGINEERING

ASSESSMENT TOOL 2 – DATA ANALYSIS PROJECT

Course: Data Handling and Data Visualization

Code: DSA0601

CO: CO3

Duration: 120 min

Total Marks: 100

Reg No: 192424375

Name: C Nithisha

COURSE OUTCOME COVERED

CO3: Analyze time-series data, trends, associations, and uncertainty through appropriate visualization methods. (BL4)

Activity Tool : Data Analysis Project

Weightage: 20%

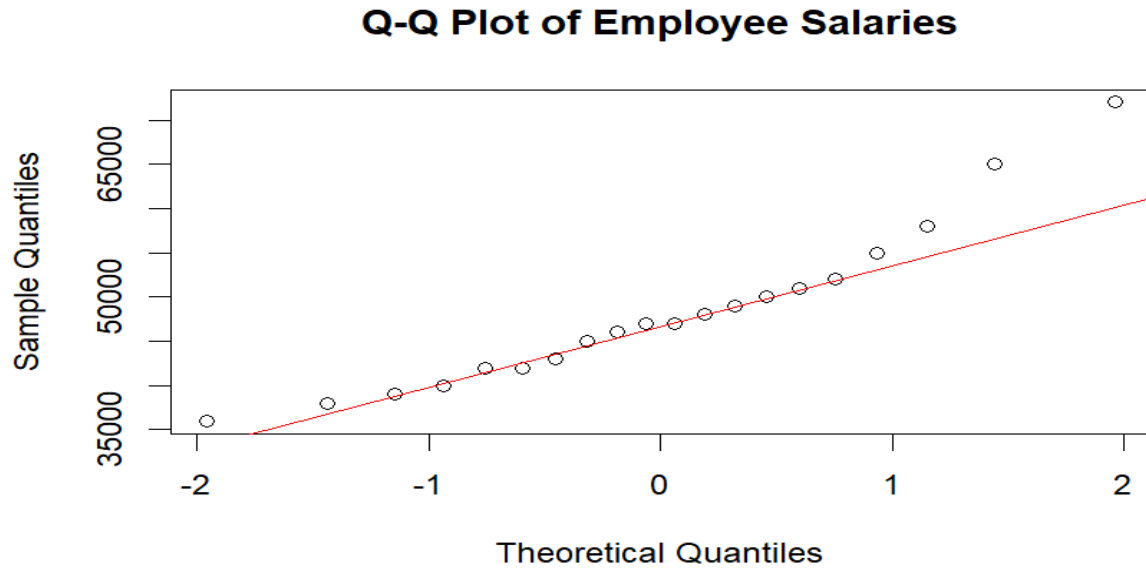
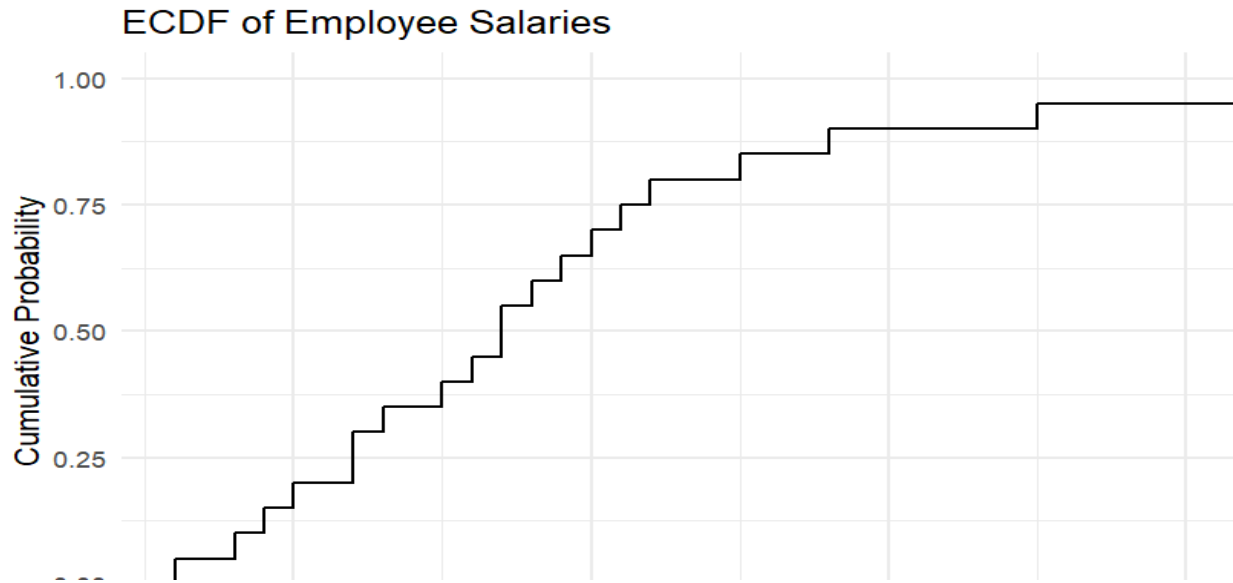
Code of Conduct:

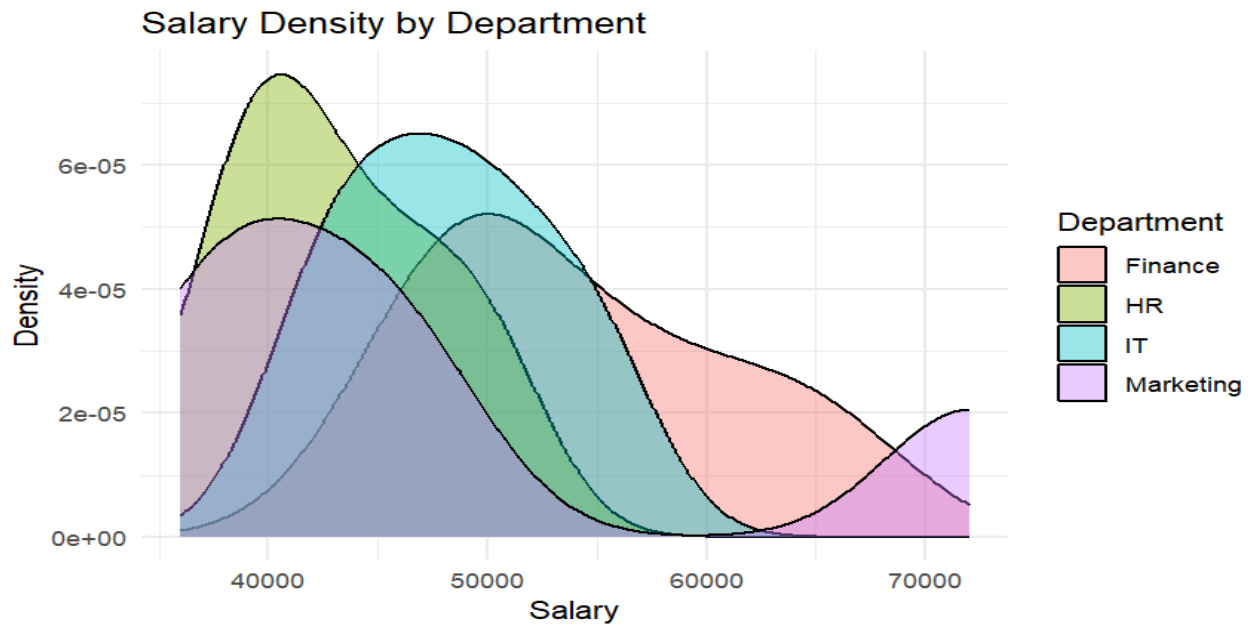
I **C Nithisha (192424375)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **C Nithisha**

Criteria	Problem Understanding & Data Preparation 25%	Data Analysis & Interpretation 30%	Application of Analytical Techniques 20%	Analysis of Results 15%	Organization & Documentation 10%	Presentation & Clarity 5%	Total
Q1							
Q2							
Q3							
Q4							
Q5							

QUESTION-1 ECDF, Q-Q Plot and Department Salary Distribution





CODE:

```
library(ggplot2)
library(dplyr)
# Create dataset
salary <- data.frame(
  Employee_ID = paste0("E", sprintf("%03d", 1:20)),
  Department = c(
    rep("IT", 5),
    rep("HR", 5),
    rep("Finance", 5),
    rep("Marketing", 5)
  )
)
```

```

rep("Marketing", 5) ),
Salary = c(
  42000, 45000, 48000, 51000, 55000,
  38000, 40000, 42000, 46000, 50000,
  47000, 49000, 52000, 58000, 65000,
  36000, 39000, 43000, 47000, 72000))
print(salary)
ggplot(salary, aes(x = Salary)) +
  stat_ecdf() +
  labs(
    title = "ECDF of Employee Salaries",
    x = "Salary",
    y = "Cumulative Probability" ) +
  theme_minimal()
qqnorm(
  salary$Salary,
  main = "Q-Q Plot of Employee Salaries")
qqline(
  salary$Salary,
  col = "red")
ggplot(salary, aes(x = Department, y = Salary)) +
  geom_boxplot() +
  labs(
    title = "Salary Distribution by Department",
    x = "Department",
    y = "Salary" ) +
  theme_minimal()
ggplot(salary, aes(x = Salary, fill = Department)) +
  geom_density(alpha = 0.4) +
  labs(
    title = "Salary Density by Department",
    x = "Salary",
    y = "Density" ) +
  theme_minimal()

```

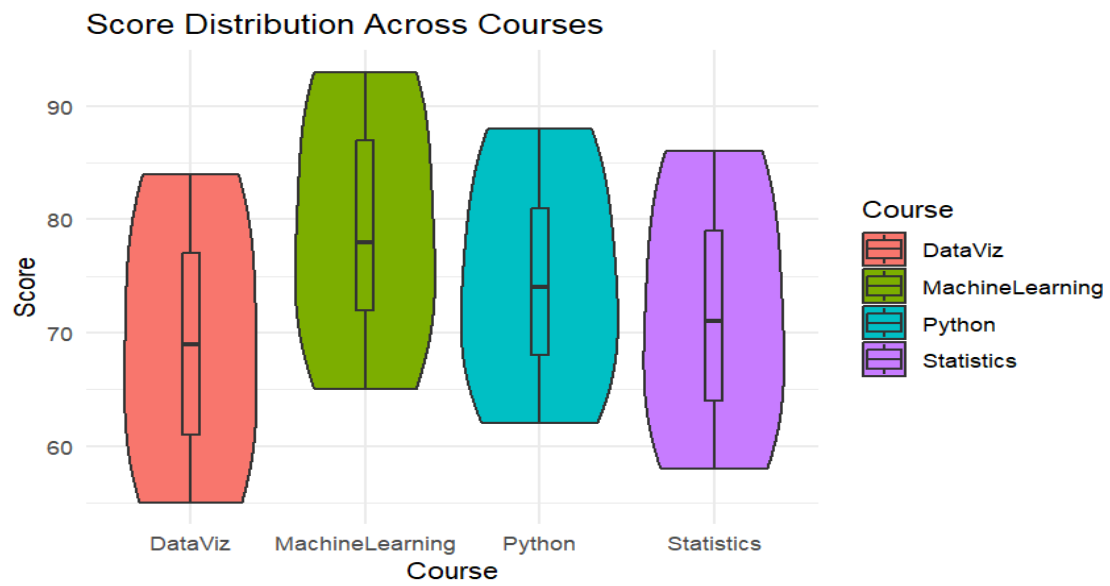
Interpretation

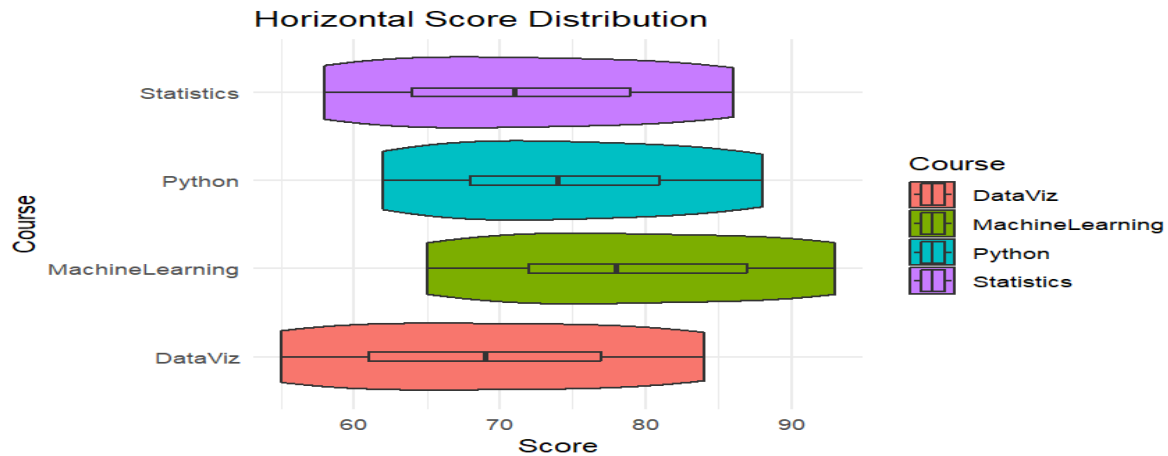
- The **ECDF** shows the cumulative proportion of employees earning below each salary level.
- The **Q-Q plot** checks whether salaries follow a normal distribution.
- Points that move away from the straight line indicate deviation from normality.
- **Marketing** has an unusually high salary of **72,000**, which may contribute to right skewness.
- **Finance** also has relatively high salaries, especially 65,000.
- Marketing has the largest salary range.

Important values

Department	Minimum	Maximum
IT	42,000	55,000
HR	38,000	50,000
Finance	47,000	65,000
Marketing	36,000	72,000

QUESTION 2 – Bean/Violin Plots





CODE:

```
library(ggplot2)
# Create dataset
student <- data.frame(
  Student_ID = paste0("S", sprintf("%03d", 1:20)),
  Course = c(
    rep("Python", 5),
    rep("Statistics", 5),
    rep("DataViz", 5),
    rep("MachineLearning", 5)),
  Study_Hours = c(
    5, 7, 9, 11, 14,
    4, 6, 8, 10, 13,
    3, 5, 7, 9, 12,
    6, 8, 10, 13, 16 ),
  Score = c(
    62, 68, 74, 81, 88,
    58, 64, 71, 79, 86,
    55, 61, 69, 77, 84,
    65, 72, 78, 87, 93 ))
# View data
print(student)
ggplot(student, aes(x = Course, y = Score, fill = Course)) +
  geom_violin() +
  geom_boxplot(width = 0.1) +
  labs(
    title = "Score Distribution Across Courses",
    x = "Course",
    y = "Score"
```

```

) +
theme_minimal()
ggplot(student, aes(x = Score, y = Course, fill = Course)) +
  geom_violin() +
  geom_boxplot(width = 0.1) +
  labs(
    title = "Horizontal Score Distribution",
    x = "Score",
    y = "Course"
  ) +
  theme_minimal()

```

Interpretation

- The violin plot shows the **shape and spread** of scores.
- **MachineLearning** has the highest scores overall.
- MachineLearning scores range from **65 to 93**.
- **DataViz** has the lowest scores overall.
- Python has a relatively high and consistent performance.
- The courses have similar score ranges, but their centers differ.

Highest-performing course

MachineLearning

Average scores:

Python = 74.6

Statistics = 71.6

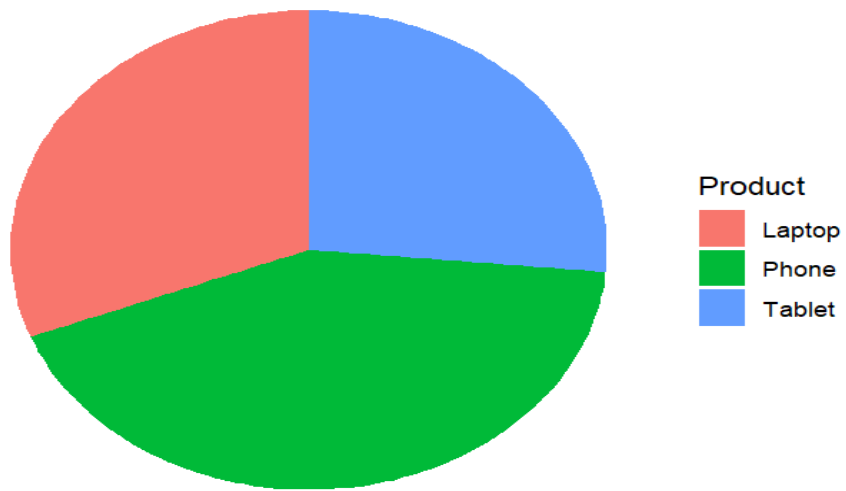
DataViz = 69.2

MachineLearning = 79.0

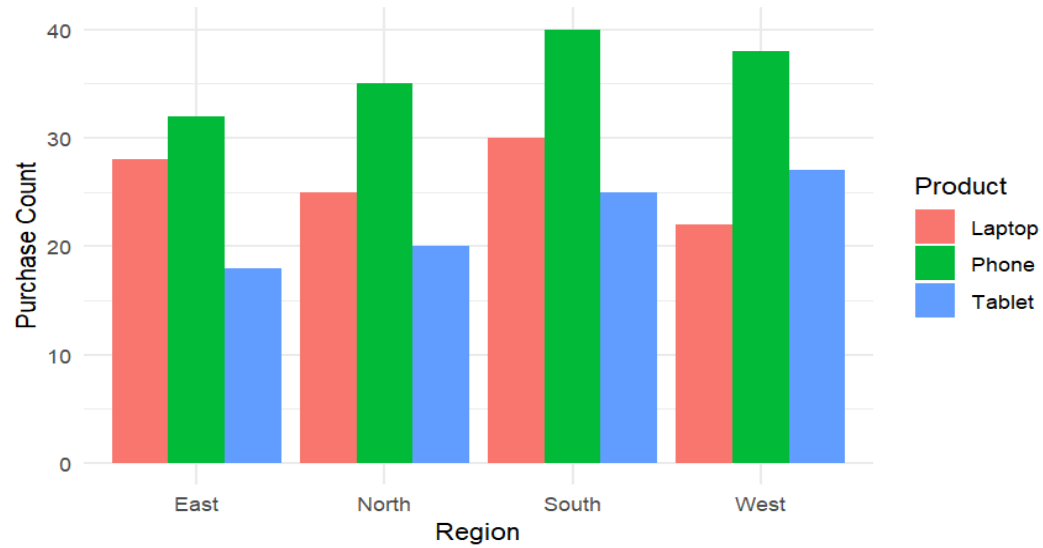
So: MachineLearning has the highest average performance.

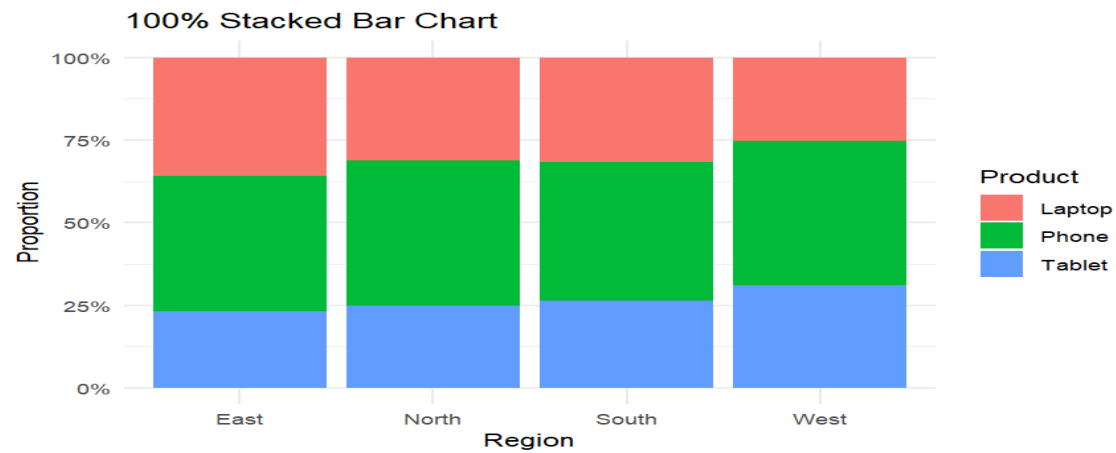
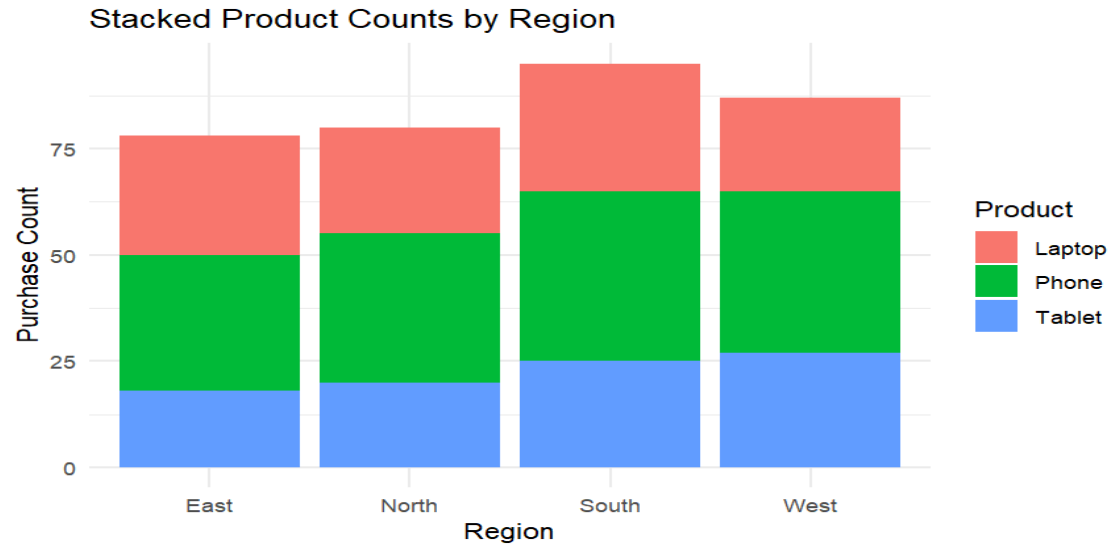
QUESTION 3 – Pie, Side-by-Side and Stacked Bar Charts

Overall Proportion of Purchases by Product



Product Counts Across Regions





CODE:

```
library(ggplot2)
```

```
library(dplyr)
```

```
# Create dataset
```

```
purchase <- data.frame(
```

```
  Customer_ID = paste0("C", sprintf("%03d", 1:12)),
```

```
  Region = c(
```

```
    "North", "North", "North",
```

```
    "South", "South", "South",
```

```

"East", "East", "East",
"West", "West", "West"),
Product = c(
  "Laptop", "Phone", "Tablet",
  "Laptop", "Phone", "Tablet",
  "Laptop", "Phone", "Tablet",
  "Laptop", "Phone", "Tablet"),
Purchase_Type = c(
  "Online", "Online", "Store",
  "Store", "Online", "Store",
  "Online", "Store", "Online",
  "Store", "Online", "Store"),
Count = c(
  25, 35, 20,
  30, 40, 25,
  28, 32, 18,
  22, 38, 27))
print(purchase)
product_total <- purchase %>%
  group_by(Product) %>%
  summarise(Total = sum(Count))
ggplot(product_total, aes(x = "", y = Total, fill = Product)) +
  geom_bar(stat = "identity", width = 1) +

```

```

coord_polar("y") +

labs(

  title = "Overall Proportion of Purchases by Product" ) +

theme_void()

ggplot(purchase, aes(x = Region, y = Count, fill = Product)) +

geom_bar(

  stat = "identity",

  position = "dodge" ) +

labs(

  title = "Product Counts Across Regions",

  x = "Region",

  y = "Purchase Count" ) +

theme_minimal()

ggplot(purchase, aes(x = Region, y = Count, fill = Product)) +

geom_bar(

  stat = "identity",

  position = "stack" ) +

labs(

  title = "Stacked Product Counts by Region",

  x = "Region",

  y = "Purchase Count" ) +

theme_minimal()

ggplot(purchase, aes(x = Region, y = Count, fill = Product)) +

```

```
geom_bar(  
  stat = "identity",  
  position = "fill"  
) +  
labs(  
  title = "100% Stacked Bar Chart",  
  x = "Region",  
  y = "Proportion"  
) +  
scale_y_continuous(labels = scales::percent) +  
theme_minimal()
```

Interpretation

The **pie chart** is useful for showing the overall proportion of products.

The **side-by-side bar chart** is better for comparing actual product counts between regions.

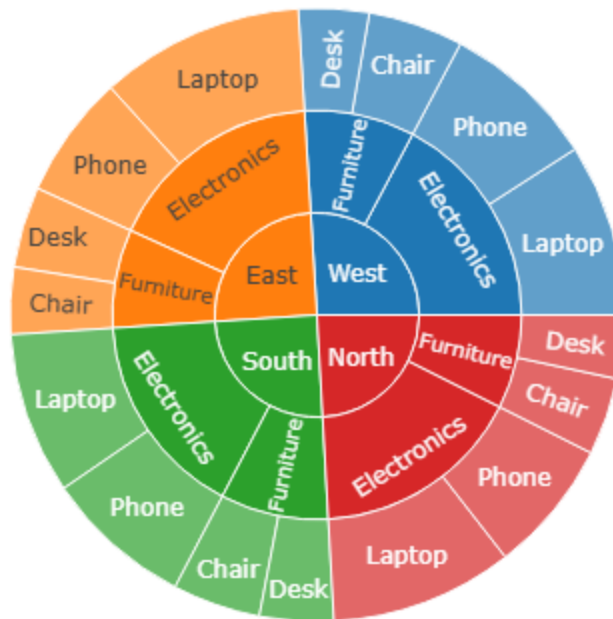
The **stacked bar chart** shows the total purchases as well as product composition.

The **100% stacked bar chart** is the best choice when the objective is to compare **regional proportions**, because every region has the same total height of 100%.

Important observation

For regional proportions: 100% stacked bar chart is more appropriate than a pie chart because it makes it easier to compare the product composition of different regions.

QUESTION 4 – Treemap and Sunburst



CODE:

```
sales <- data.frame(
```

```
  Region = c(
    "North","North","North","North",
    "South","South","South","South",
    "East","East","East","East",
    "West","West","West","West"
  ),
```

```
  Category = c(
    "Electronics","Electronics","Furniture","Furniture",
    "Electronics","Electronics","Furniture","Furniture",
    "Electronics","Electronics","Furniture","Furniture",
    "Electronics","Electronics","Furniture","Furniture"
  ),
```

```

Subcategory = c(
  "Laptop","Phone","Chair","Desk",
  "Laptop","Phone","Chair","Desk",
  "Laptop","Phone","Chair","Desk",
  "Laptop","Phone","Chair","Desk"
),

Sales = c(
  52000,38000,22000,18000,
  46000,42000,25000,21000,
  58000,35000,19000,23000,
  49000,44000,27000,20000
)
)

print(sales)

ggplot(
  sales,
  aes(
    area = Sales,
    fill = Category,
    label = Subcategory,
    subgroup = Region,
    subgroup2 = Category
  )
) +

  geom_treemap() +

  geom_treemap_subgroup_border(
    colour = "black",
    size = 2
  ) +

  geom_treemap_subgroup2_border(
    colour = "white",
    size = 1
  ) +

```

```

geom_treemap_text(
  aes(label = paste(Subcategory, "\n$", Sales)),
  colour = "black",
  place = "centre",
  grow = TRUE
) +

labs(
  title = "Company Sales Treemap",
  subtitle = "Region → Category → Subcategory",
  fill = "Category"
) +

theme_minimal()

```

Region totals

```

region <- aggregate(
  Sales ~ Region,
  data = sales,
  sum
)

```

Region + Category totals

```

category <- aggregate(
  Sales ~ Region + Category,
  data = sales,
  sum
)

```

Create Region IDs

```

region$id <- region$Region

```

```

region$parent <- ""

```

```
# Create Category IDs
```

```
category$id <- paste(  
  category$Region,  
  category$Category,  
  sep = "-"  
)
```

```
category$parent <- category$Region
```

```
# Create Subcategory IDs
```

```
subcategory <- sales
```

```
subcategory$id <- paste(  
  subcategory$Region,  
  subcategory$Category,  
  subcategory$Subcategory,  
  sep = "-"  
)
```

```
subcategory$parent <- paste(  
  subcategory$Region,  
  subcategory$Category,  
  sep = "-"  
)
```

```
# Combine all three levels
```

```
sunburst_data <- rbind(  
  
  data.frame(  
    id = region$id,  
    label = region$Region,  
    parent = region$parent,  
    value = region$Sales
```



```

),

data.frame(
  id = category$id,
  label = category$Category,
  parent = category$parent,
  value = category$Sales
),

data.frame(
  id = subcategory$id,
  label = subcategory$Subcategory,
  parent = subcategory$parent,
  value = subcategory$Sales
)
)

```

Create Sunburst

```

plot_ly(
  data = sunburst_data,
  ids = ~id,
  labels = ~label,
  parents = ~parent,
  values = ~value,
  type = "sunburst",
  branchvalues = "total"
)

```

Largest Region

```

region_sales <- aggregate(
  Sales ~ Region,
  data = sales,
  sum
)

```

```
print(region_sales)
```

```
largest_region <- region_sales[  
  which.max(region_sales$Sales),  
]
```

```
print(largest_region)
```

```
# Largest Category
```

```
category_sales <- aggregate(  
  Sales ~ Category,  
  data = sales,  
  sum  
)
```

```
print(category_sales)
```

```
largest_category <- category_sales[  
  which.max(category_sales$Sales),  
]
```

```
print(largest_category)
```

```
# Largest Subcategory
```

```
subcategory_sales <- aggregate(  
  Sales ~ Subcategory,  
  data = sales,  
  sum  
)
```

```
print(subcategory_sales)
```

```
largest_subcategory <- subcategory_sales[  
  which.max(subcategory_sales$Sales),  
]
```

```
print(largest_subcategory)
```

Interpretation

- **West is the largest region**, with total sales of **140,000**.
- **Electronics is the largest category**, with total sales of **364,000**, which is higher than Furniture.
- **Laptop is the largest subcategory**, with total sales of **205,000**.
- The **treemap** is easier for comparing sales sizes, while the **sunburst chart** is better for showing the hierarchy of *Region → Category → Subcategory*.
- A limitation of the **treemap** is that small areas can be difficult to read, while the **sunburst** can become crowded when there are many categories.

Highest-Selling Region

West

Average/Total Sales:

North = 130,000

South = 134,000

East = 135,000

West = 140,000

So: **West has the highest total sales.**

Highest-Selling Category

Electronics

Total Sales:

Electronics = 364,000

Furniture = 175,000

So: **Electronics has the highest total sales.**

Highest-Selling Subcategory

Laptop

Total Sales:

Laptop = 205,000

Phone = 159,000

Chair = 93,000

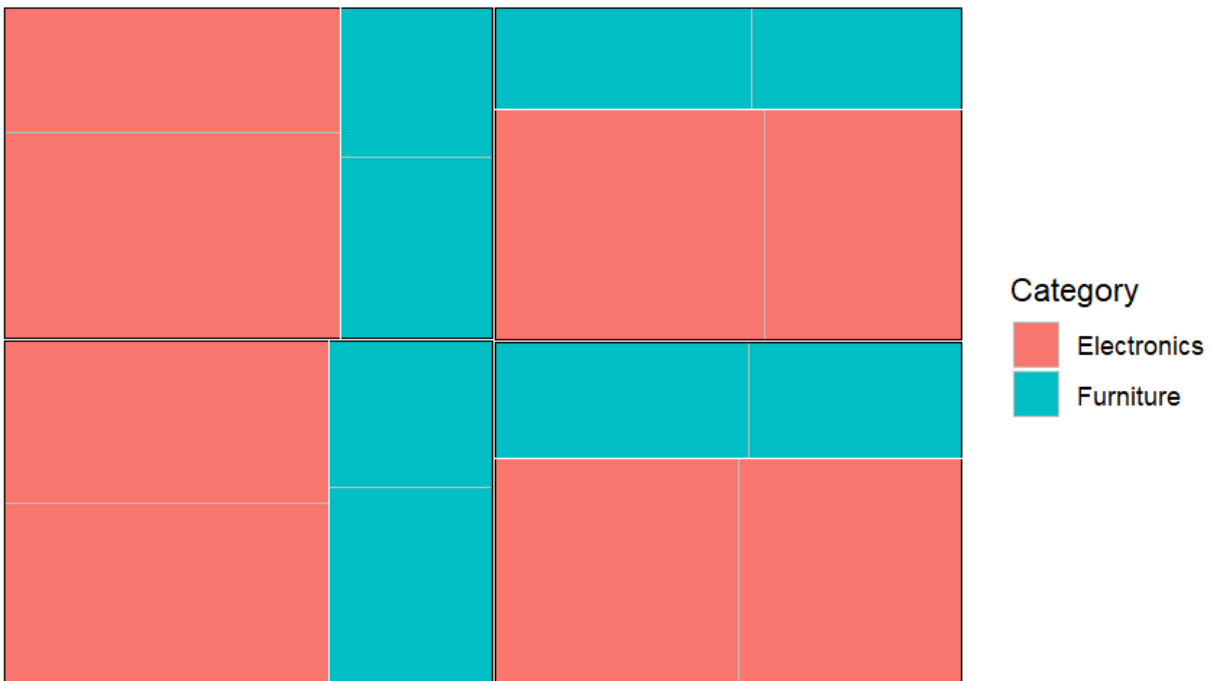
Desk = 82,000

So: Laptop has the highest total sales

Question 5 – Flow-Based Proportions with Sankey Diagram and Parallel Sets

Company Sales Treemap

Region → Category → Subcategory



CODE:

```
install.packages("plotly")
install.packages("ggplot2")
install.packages("ggalluvial")
```

```
library(plotly)
library(ggplot2)
library(ggalluvial)
```

```
career <- data.frame(
```

```
  Student_Group = c(
    "G1","G2","G3",
    "G4","G5","G6",
    "G7","G8","G9",
    "G10","G11","G12"
  ),
```

```
Specialization = c(
  "Data Analytics",
  "Data Analytics",
  "Data Analytics",
  "AI",
  "AI",
  "AI",
  "Cyber Security",
  "Cyber Security",
  "Cyber Security",
  "Cloud Computing",
  "Cloud Computing",
  "Cloud Computing"
),
```

```
Internship = c(
  "Yes","Yes","No",
  "Yes","Yes","No",
  "Yes","Yes","No",
  "Yes","Yes","No"
),
```

```
Placement_Sector = c(
  "IT",
  "Finance",
  "IT",
  "IT",
  "Healthcare",
  "IT",
  "IT",
  "Finance",
  "Government",
  "IT",
  "Finance",
  "Government"
),
```

```
Students = c(
  18,12,8,
  20,10,6,
```

```
    16,9,7,  
    15,8,5  
  )  
)
```

```
print(career)
```

```
# Create nodes
```

```
nodes <- data.frame(  
  name = c(  
    "Data Analytics",  
    "AI",  
    "Cyber Security",  
    "Cloud Computing",  
    "Yes",  
    "No",  
    "IT",  
    "Finance",  
    "Healthcare",  
    "Government"  
  )  
)
```

```
# Find source and target positions
```

```
source1 <- match(  
  career$Specialization,  
  nodes$name  
) - 1
```

```
target1 <- match(  
  career$Internship,  
  nodes$name  
) - 1
```

```
source2 <- match(  
  career$Internship,
```

```

    nodes$name
  ) - 1

target2 <- match(
  career$Placement_Sector,
  nodes$name
) - 1

# Create links

links <- data.frame(
  source = c(source1, source2),
  target = c(target1, target2),
  value = c(career$Students, career$Students)
)

# Create Sankey diagram

fig <- plot_ly(
  type = "sankey",
  orientation = "h",

  node = list(
    label = nodes$name,
    pad = 15,
    thickness = 20
  ),

  link = list(
    source = links$source,
    target = links$target,
    value = links$value
  )
)

fig <- fig %>%
  layout(
    title = "Student Career Path Sankey Diagram"
  )

```



```
)
```

```
fig
```

```
ggplot(  
  career,  
  aes(  
    axis1 = Specialization,  
    axis2 = Internship,  
    axis3 = Placement_Sector,  
    y = Students  
  )  
) +
```

```
  geom_alluvium(  
    aes(fill = Specialization),  
    alpha = 0.7  
  ) +
```

```
  geom_stratum() +
```

```
  geom_text(  
    stat = "stratum",  
    aes(label = after_stat(stratum))  
  ) +
```

```
  scale_x_discrete(  
    limits = c(  
      "Specialization",  
      "Internship",  
      "Placement Sector"  
    )  
  ) +
```

```
  labs(  
    title = "Parallel Sets of Student Career Paths",  
    x = "Career Path",  
    y = "Number of Students",  
    fill = "Specialization"  
  ) +
```

```
theme_minimal()

internship_total <- aggregate(
  Students ~ Internship,
  data = career,
  sum
)

print(internship_total)


placement_total <- aggregate(
  Students ~ Placement_Sector,
  data = career,
  sum
)

print(placement_total)


specialization_total <- aggregate(
  Students ~ Specialization,
  data = career,
  sum
)

print(specialization_total)
```

Interpretation

1. *AI → Yes → IT is the largest individual flow with 20 students.*
2. Students with internships (108) are much higher than students without internships (26).
3. IT is the most common placement sector with 83 students.
4. The Sankey diagram communicates flow structure more clearly because wider bands represent more students.
5. Incorrect scales, colors, or labels can make a small flow look larger than it actually is and lead to a misleading conclusion.

Internship

Yes = 108

No = 26

Placement Sector

IT = 83

Finance = 39

Healthcare = 10

Government = 12

Specialization

Data Analytics = 38

AI = 36

Cyber Security = 32

Cloud Computing = 28

Major Flow

The largest individual flow is:

AI → Yes → IT = 20 students

Other major flows:

- *Data Analytics* → Yes → IT = 18
- *Cyber Security* → Yes → IT = 16
- *Cloud Computing* → Yes → IT = 15